
Local Management Instructions

LONG RETRACTABLE SOOT BLOWER OPERATION



Approved for implementation by HEAD OF PROJECT

Date 11.12.2022

Prepared by OPERATION DEPARTMENT

NTPC LTD.

Simhadri Super Thermal Power Project.

Long Retractable Soot Blower Operation
LMI/OGN/OPS/SYST/005 Issue 01: Rev-06 Dec 2022

LONG RETRACTABLE SOOT BLOWER OPERATION

LMI	LONG RETRACTABLE SOOT BLOWER OPERATION		
LMI No	LMI/OGN/OPS/SYST/005		
Issue No	1	Revision: 6	Dt: 11.12.2022

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1	INTRODUCTION
	One of the most important boiler auxiliary operation is the on-line fire-side cleaning of heat-absorbing surfaces. Soot blowing is important not only to ensure proper heat transfer, but also to prevent section of the boiler from becoming severely blocked. Blocked sections can restrict gas flow, cause tube erosion due to high local gas velocities and result in load limitations. Also, soot blowing becomes important when burning fuels other than those for which the boiler is designed. Further, Soot blowing is critical in maintaining the proper balance of heat transfer in the boiler and matching steam rate with the superheat and reheat capability at any given load. Various types of Soot Blowers used in Boiler are: 1. Wall Soot Blowers/ Deslagers. 2. Long Retractable Soot Blowers (LRSB) & Half Retractable Soot Blower (HRSB) 3. Soot blowers for Air preheaters/ Air preheater Cleaning Device While wall blowers find application in furnace wall cleaning, Long Retractable Soot Blowers (LRSBs) are used to clean the superheater, reheater, and sometimes the economiser passes. The Air Preheater soot blowers are used for cleaning air preheater during normal operation by blowing fly ash that has been deposited by the passing flue gas. Improper and inadequate attention and management of soot blower operation cause boiler tube erosion and can lead to avoidable tube failures in boiler pressure parts. Soot blower steam temperature degree of superheat is one of the most critical parameter to be maintained. It is recommended to have 150 to 165 C superheat of steam for steam blowing.
2	SUPERSEDED DOCUMENTS
	Superseded - LMI/OGN/OPS/SYST/005 Issue-1, Rev-05 Dated: MAY'2022
3	RECORDS OF REVISION NOTES
	i. Clause 1.0: Introduction elaborated, recommendation for Degree of Superheat of soot-blowing steam has been incorporated. ii. Section 5: Design: elaborated iii. Section 5.1 Installation- elaborated iv. Clause 6.2.5: Soot blowers and frequency of operation – elaborated v. Clause 7.4: Emergency operation of soot blowers - elaborated. vi. Clause 8.0: Record for soot blowing line drain temperature and degree of superheat incorporated. vii. Design check on soot blowers and its appendix-I have been removed as it is not required in operating units.
4	SCOPE
	This LMI provides guidance on certain aspects of soot blower design and installation, the frequency of LRSB operation, emergency operation on failure of power or control supply, blowing medium or when the lance become stalled. Some of the observations relating to boiler parameters before and after LRSB operation are indicated. This LMI is applicable to Stage-1&2 units of Simhadri.

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5	SOOT BLOWER DESIGN & INSTALLATION
	The efficiency of the entire heat transfer cycle is determined by the ability to maintain the cleanliness of furnace. A severely slagged furnace causes fouling in other components such as the SH, RH, Economizer, thus requiring higher than normal furnace exit gas temperatures to meet original steam design temperatures. LRSB are used for cleaning ash deposition in convective surfaces of SH, RH and sometimes economisers. The LRSB uses two cleaning nozzles at the tip, 180degree opposite to each other and emit a high-energy jet of superheated steam perpendicular to the lance. While the lance traverses the boiler, it rotates, forming a helical blowing pattern thereby effectively cleans the tubes and spaces between tubes in a superheater, reheater, or economizer bank. To improve the cleaning effect and to reduce erosions, the soot blower is retracted in a different helical path (indexed) each time the blower is operated. The Lance operating in the high gas temperature zones of the boiler are made of 2.25% Cr/1% Mo or 1% Cr/1/2% Mo steel. These materials have inadequate strength when subjected to the temperature above 600°C and 12% Cr. steel is often used where the temperatures exceed 600 C. The total stress on a long travel lance is the static bending stress plus a dynamic component due to lance rotation. Further Large residual deposits of slag present in the upper areas of the boiler (after the boiler cools down) would present hazards to maintenance personnel if left in place. Soot blowers and LRSBs are to be operated during normal boiler load seeing the flame condition. However, they can be operated offload also if unit does not have high unburnt in fly ash. This will help in cleaner and safer boiler for inspection. Will help in removal of dead ash weight inside boiler to facilitate faster cooling.

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5.1	INSTALLATION
	1) It is essential to maintain the predetermined slope in the cold condition as given in the erection drawings when installing LRSBs. The lance droop, in the fully extended condition, must be maintained as per the manufacturer's recommendations. This aspect must be checked and recorded. Maintenance of correct slope in the cold condition (as suggested by OEM) will ensure proper draining of the steam lines and avoid blowing condensate on the tubes and consequently any premature tube failure. 2) The clearance between the boiler tubes and lance must be maintained as per the manufacturer's recommendations. This check should be carried out by operating the blower manually. The clearance should consider boiler expansion in hot conditions. These minimum checks should also be carried out during overhauls and extended outages and are essential to avoid any damage due to steam/condensate impingement. The high velocity condensate through soot blower nozzles can cause severe boiler tube erosion, water hammer in soot blower pipes, damage to soot blower nozzle head & valve seat etc.
	Responsibility - In charge (BM-PP)
6.0	CONVECTION SURFACE LRSBs AND FREQUENCY OF OPERATION
	The long retractable type soot blower is the most effective way to clean radiant and convective heating surfaces. The type of deposits can vary from very hard tenacious slag to a powdery coating. The blowing sequence and frequency of the retract system must be optimized during initial operation by starting with an assumed sequence of blowing and a frequency of, say, one complete cycle per week. The Head of Operation may modify the blowing pattern and the frequency after observing the fouling patterns in the boiler exit gas temperature and increase in gas pressure across the boiler (assuring boiler at constant load). It is common for certain LRSBs to be blown more frequently than others, either singly or in groups. Once the pattern is established, it can be programmed in the soot blower system and run automatically. Utmost care must be exercised by the operating staff during soot blowing cycle even in automatic sequence to ensure complete withdrawal of lance after retraction. It is not acceptable however, for the LRSBs to remain unused for long intervals of time.
	Responsibility: SCE
6.1	FREQUENCY OF LRSB OPERATION
	The frequency of LRSB operation depends on the operating conditions of the Boiler & Boiler cleanliness. Some parameters which may indicate the need for LRSB Operation are a) Increase in APH flue gas inlet temperature b) Flue gas Temperature profile across various pressure parts like SH, RH, LTSH coils (at boiler constant load) c) Visual inspection of Boiler furnace and pressure parts. Following is the schedule of LRSB operation carried out in unit # 1, 2, 3 & 4 based on the

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	dynamic condition of the boiler. However, this may be revised as per the requirement of the optimization and other factors, which will be intimated to shift operation time to time.						
	DAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
	UNIT	UNIT#1	UNIT#2	UNIT#3	UNIT#4	UNIT#1&2	UNIT#3&4
	NO OF LRSBS OPERATED	12	12	12	12	12+12	12+12
6.2	INSTRUCTIONS:						
	Instruct the operators to check the following:				Responsibility - Unit Controller		
	a) Lubricating oil levels for gear boxes						
	b) Ensure the availability of hand cranks before operating the blowers						
	c)Ensure the blower lances are in home position before and also after the soot blowing cycle is completed.						
	d)Ensure the valve opening linkages are in good working condition and operation is smooth.						
	e) Ensure the manual isolating valves are open before starting of soot blowing cycle and close after shoot blowing cycle.						
	f) Ensure that steam line drain control valve is line up condition and it is operating in auto as per the temperature set point						
	g) Ensure the availability of maintenance staff (MMD, EMD) along with operating handles, tools and fuses before start of soot blowing cycle. LWC to be given to EMD for LRSB operation assistance.						
	h) Ensure the availability of communication system between the Boiler operating desk engineer and operator.						
	i)Operating chains and lance tubes are to be checked for any foreign material stuck up before the start of blower.						
	j) Ensure steam flow low permissive for blower operation.						
6.2.1	a) Ensure no alarm is persisting in Soot blowing panel. b) Select Manual start from Soot Blowing Panel.						
6.2.2	a) Open the MOV in steam line and crack open the control valve to warm up the steam line. b) After attaining the temperature (250 °C) keep the control valve in auto and set point at 25 ksc.						
6.2.3	Ensure the interlocks are in operating condition before the start of each blower Operation. A) Home position limit switch operation. b) Advance limit switch operation						
6.2.4	As the start push button is pressed, the unit begins to extend and rotate. As the nozzles clear the Boiler wall, the carriage rotates the 'V' shaped liner opening the head and blowing commences. At fully extended position the travelling carriage strikes the reverse limit switch, which closes momentarily to reverse the unit. As the carriage retracts, it again strikes the 'V' shaped liner closing SBV head. The travelling carriage continues it retract travel until the cleaning nozzle is retracted into the protection of wall box at which time the stop limit switch is stuck to stop the unit.						
6.2.5	Utmost care must be exercised by the operating staff during soot blowing cycle even in automatic sequence to ensure complete withdrawal of lance after retraction.						

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7.0	EMERGENCY OPERATION OF LRSBS
7.1	Loss of blowing medium when the lance is inserted results in a rapid increase in the lance temperature and unless it is retracted immediately, mechanical failure often results. If a blower stops during its cleaning cycle, it must be retracted immediately. A static lance supplied with steam will erode any tube in the path of the jets, since the jets entrain gas and dust from the surrounding gas. During retraction manually it must be tried to rotate the same so as to cause minimum damage to lance and coil tubes.
	Responsibility – SCE/In charge (BM-PP)/In charge (EMD)
7.2	If the power or control supply fails, the blower should be immediately brought to its starting position manually to avoid any possible damage to heating surfaces or the lance tube. It is thus necessary to retract LRSBs manually in case of a) Power/control supply failure b) Over run of LRSBs due to defective limit switches/chain failure or other electrical faults
	Responsibility – SCE/In charge (BM-PP)/In charge (EMD)
7.3	For this purpose, sufficient number of hand cranks should be readily available. Before attempting manual retraction with hand cranks, steam lines and power supply to LRSB motor must be quickly isolated.
	Responsibility – SCE/In charge (BM-PP)/In charge (EMD)
7.4	GM (Maintenance)/ GM (O&M) must ensure the availability of hand cranks and also ensure a dedicated group of operators trained on LRSB operation and maintenance. This group shall be fully familiar with the procedure of LRSB manual operation, as per manufacturer's instructions. The operators must be fully aware that soot blowers should never be manually operated without the steam supply isolated.
	Responsibility – SCE/In charge / (BM-PP)/In charge (EMD)/ GM (Maintenance)/ GM (O&M)
	Additional care to be taken is: Responsibility-UC
	No LRSB should be operated for a second time within one-hour interval to avoid damage to motor windings. LRSB should never be manually operated (advance and retraction) with steam When alarm is sounded due to I) Blower over run, overload II) Steam pressure low III) Steam flow low IV) Interlocks, etc. Ensure that the lance is withdrawn immediately.
7.5	TO PUT THE LRSB ON MANUAL OPERATION (MANUAL RETRACTION)
	a) Switch off the power and control supply for the blower. b) Never attempt to delink the chain while blowing steam is charged condition as the force in the lance would drive the unit against the lance and yoke bearing plate assembly. c) Insert the hand crank and rotate until the blower reaches home position. d) Remove the hand crank at blower home position.
	Responsibility – SCE/In charge (BM-PP)/In charge (EMD)

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8.0	RECORDS TO BE MAINTAINED FOR LRSB OPERATION
	The LRSB operation records shall be maintained by Unit Operator. These records should show as a minimum the following: - • Time allowed for draining the entire soot blower system • The date and time of operation of LRSBs with blower numbers • Reasons for non-operation of any particular blower • Prescribed blowing pressure and • The actual blowing pressures • Status of soot blowing line drain temperature control • Degree of superheat of soot blowing steam The records should give details of the testing and operation of each individual blower with blowing medium using the local start button during annual boiler maintenance. The blowing Pressure should be recorded using a pressure gauge in the soot blower valve operating head.
9.0	RESPONSIBILITY CENTER
	Unit Control Engineer and Boiler Desk Engineer. The records pertaining to LRSB operation should be reviewed by GM (O&M) at least once every month.
10.	REVIEW
	HOD (O&M) will be responsible for reviewing this LMI on two yearly basis or as necessary

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11.0 TECHNICAL SPECIFICATIONS:

TYPE: Fully Automatic sequentially controlled processor (PR05) based steam soot blowing

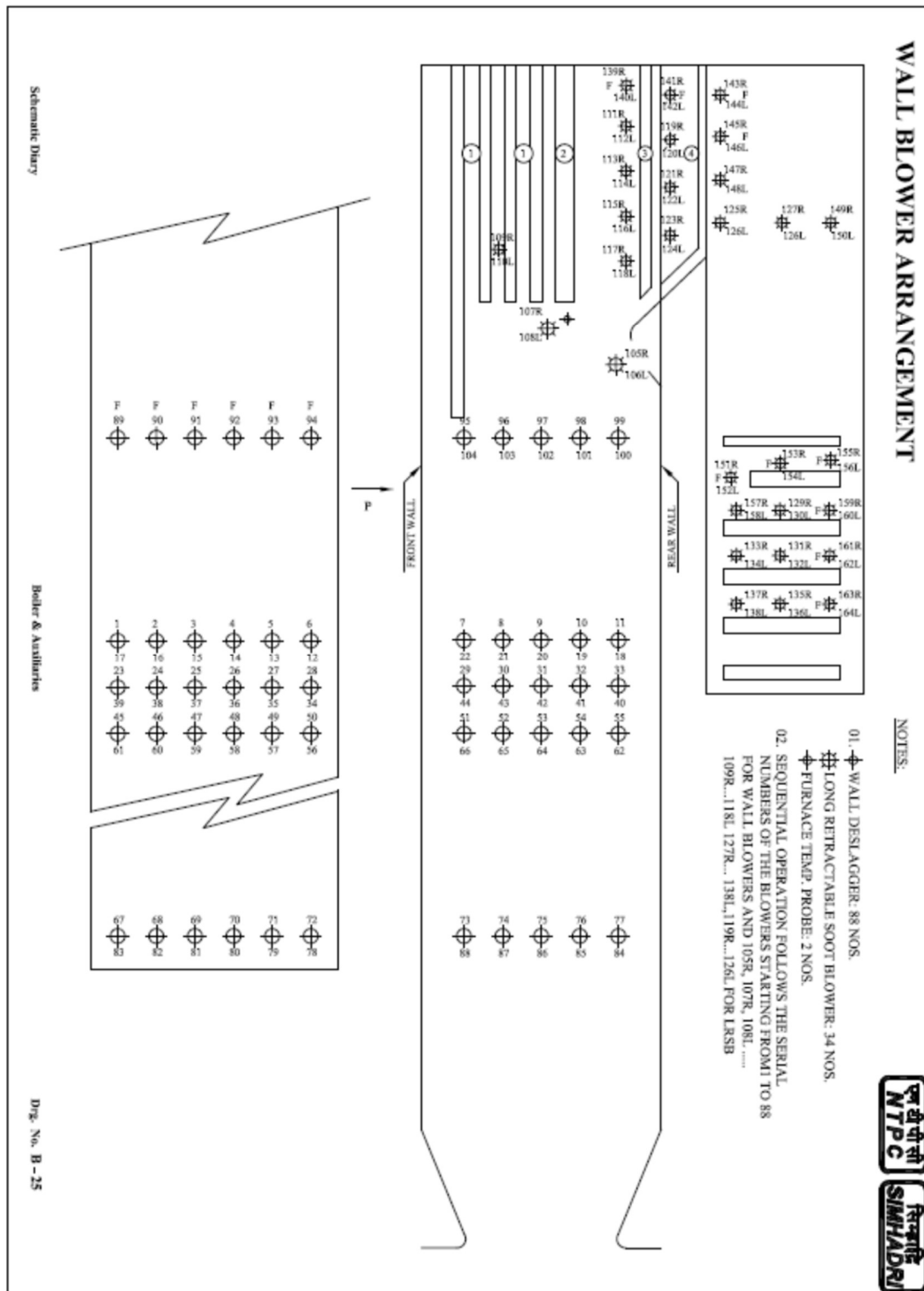
STAGE#1

SOURCE OF STEAM	LTSH OUTLET HEADER
a) Number of blowers	34(105-138)
b) Type	LRD II E
c) Travel in mm	9700 mm
d) Nozzle size / number	32/2(105-108),25/2(109-138)
e) Blowing pressure	12.5ksc (105-124),10 ksc(125-138)
f) Operating time	12.5 min/blower cycle

STAGE#2

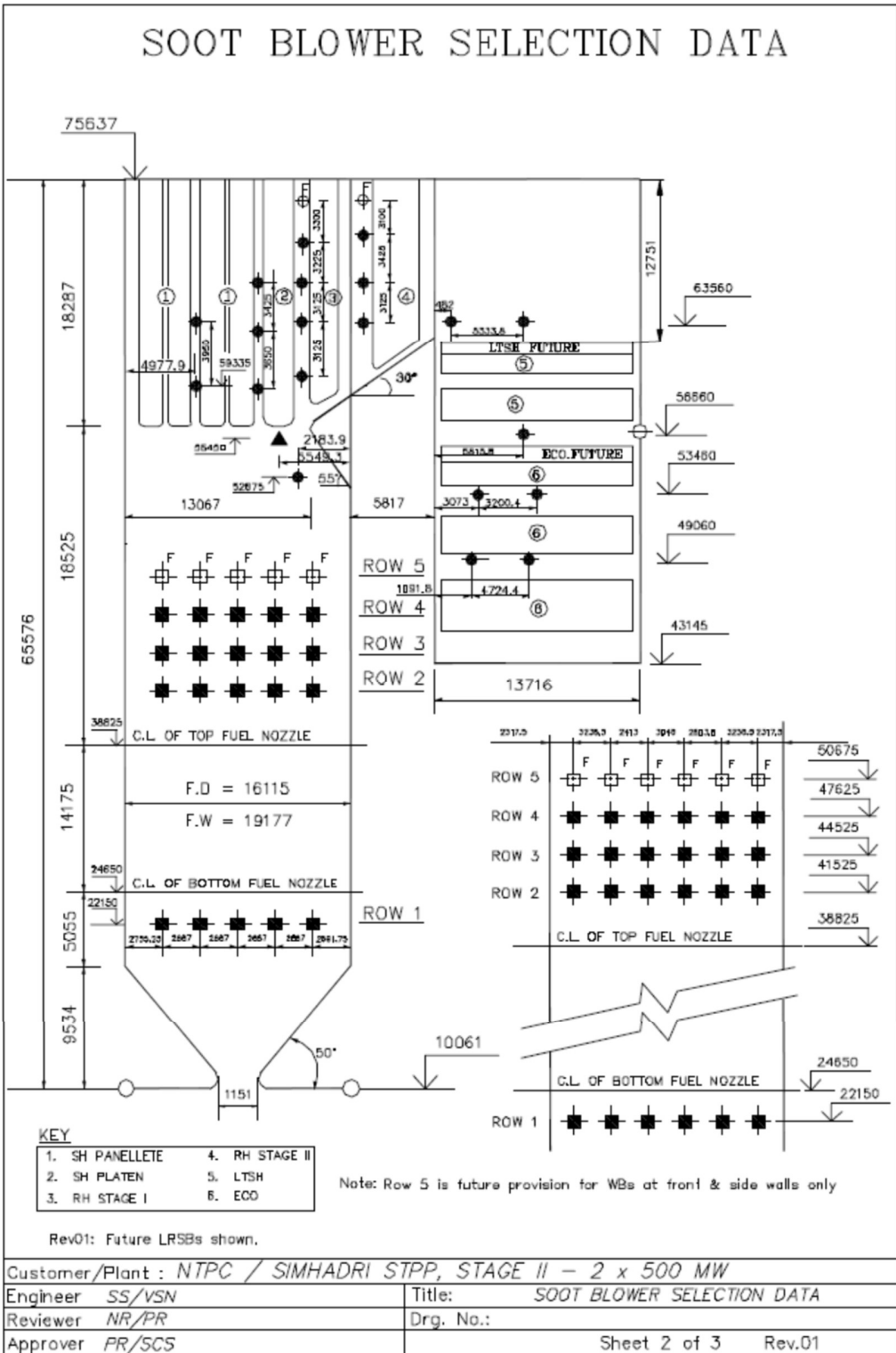
SOURCE OF STEAM	LTSH OUTLET HEADER
a) Number of blowers	40 (105-144)
b) Type	LRD II E
c) Travel in mm	9700 mm
d) Nozzle size / number	<u>32/2</u> (105,106,109,110,115,116),- <u>25/2</u> (107,108,111-114,117-144)
e) Blowing pressure	12.5ksc(105-124&133-144),10 .5ksc(125-132)
f) Operating time	12.5 min/blower cycle

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Stage1 Soot Blower Arrangement

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Stage2 Soot Blower Arrangement



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/641558

Date:

01/12/2022(17:17:02)

Subject: Review of LMI “GUIDE LINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER” with Ref No: LMI/OIN/OPS/MECH/032

REF.: SMPP/O&M/MM/PP/

Date: 01.12.2022

Subject: Revision of LMI “GUIDE LINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER” with Ref No: LMI/OIN/OPS/MECH/032

-

The above approved LMI was revised in month of Aug '2022. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

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On: 01/12/2022(17:17:02) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/641558

The subject LMI has been revised as per changes envisaged in COS-ISO-00-OGN/OPS/MECH/053.

The changes are incorporated in LMI doc with ref no: LMI/OGN/OPS/MECH/053
ISSUE NO.1; REVISION NO. 02; DATE: DEC'2022.

CA may please accord approval for this revised version of LMI.

Submitted by:

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On:13/12/2022(16:00:57) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/641558

Changes incorporated in revised LMI in following clauses:

1. Clause 7 (A)- 2,3,4,5
2. Clause 7 (B)- 2,6,12,15,16
3. Clause 7 (C) - 2,3,5,9,10
4. Clause 7 (E)- 3

The above changes have been highlighted in attached document "New LMI doc"

Submitted for approval

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On:16/12/2022(17:29:25) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/641558

Submitted for approval pls

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May please be accorded approval.

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On:20/12/2022(15:40:31) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/641558

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On:20/12/2022(22:47:11) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/641558